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Improved brain tumor classification through DenseNet121 based transfer learning

Mehwish Rasheed¹, Muhammad Arfan Jaffar¹, Arslan Akram^{1,2,3}, Javed Rashid^{3,4}, Tagrid Abdullah N. Alshalali⁵, Asma Irshad⁶ and Nadeem Sarwar^{7*}

*Correspondence:

Nadeem Sarwar
nadeem_srwr@yahoo.com
¹Faculty of Computer Science
and Information Technology, The
Superior University, Lahore
54600, Pakistan
²Department of Computer Science,
University of People, Pasadena,
CA 91101, USA
³MLC Lab, Maharban House, House
209, Zafar Colony, Okara
56300, Pakistan
⁴Information Technology Services,
University of Okara, Okara
56300, Pakistan
⁵Department of Information
Systems, College of Computer
and Information Sciences, Princess
Nourah bint Abdulrahman
University, P.O. Box 84428,
Riyadh 11671, Saudi Arabia
⁶School of Biochemistry and
Biotechnology, University of the
Punjab, Lahore, Pakistan
⁷Department of Computer Science,
Bahria University Lahore Campus,
Lahore 54000, Pakistan

Abstract

Brain tumors have a big effect on a person's health by letting abnormal cells grow unchecked in the brain. This means that early and correct diagnosis is very important for effective treatment. Many of the current diagnostic methods are time-consuming, rely primarily on hand interpretation, and frequently yield unsatisfactory results. This work finds brain tumors in MRI data using DenseNet121 architecture with transfer learning. Model training made use of the Kaggle dataset. By preprocessing the stage, resizing the MRI pictures to minimize noise would help the model perform better. From one MRI scan, the proposed approach divides brain tissues into four groups: benign tumors, gliomas, meningiomas, and pituitary gland malignancies. The designed DenseNet121 architecture precisely classifies brain cancers. We assessed the models' performance in terms of accuracy, precision, recall, and F1-score. The suggested approach proved successful in the multi-class categorization of brain tumors since it attained an average accuracy improvement of 96.90%. Unlike previous diagnostic techniques, such as eye inspection and other machine learning models, the proposed DenseNet121-based approach is more accurate, takes less time to analyze, and requires less human input. Although the automated method ensures consistent and predictable results, human error sometimes causes more unpredictability in conventional methods. Based on MRI-based detection and transfer learning, this paper proposes an automated method for the classification of brain cancers. The method improves the precision and speed of brain tumor diagnosis, which benefits both MRI-based classification research and clinical use. The development of deep-learning models may even further improve tumor identification and prognosis prediction.

Keywords Multiclass brain tumor classification, Transfer learning, Deep learning, DenseNet121

1 Introduction

The anatomy of the brain is one of the most intricate and detailed aspects of the human body. Understanding its functions is challenging, as it involves the coordinated activity of numerous nerves. Additionally, brain-related diseases are increasingly prevalent [1]. Brain tumors are a kind of brain-related disease that may cause mortality; while



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early detection is possible for some disorders, accurate diagnosis is necessary for others. Two main categories exist within it: benign or low-level tumors and malignant or high-grade tumors [2]. High-grade tumors are malignant and can metastasize to other organs, in contrast to low-grade tumors, which are benign and do not spread. Because of their aggressive nature, high-grade tumors pose the biggest threat to a patient's life of all tumor kinds. They pose a possible hazard to human life because of their ability to spread. A variety of tumors, including pituitary tumors, meningiomas, and gliomas, can be either benign or malignant [3]. Studies and statistical reports suggest that more than 1.4 million people will be affected by brain tumors in 2023. Annually, these tumors claim the lives of approximately 18,990 individuals, including 11,020 men and 7,970 women. Projections indicate that by 2035, this number could rise to 24 million. According to Cancer Net, the survival rate for brain tumors stands at 36% over five years and approximately 30% over ten years [4].

The area of computer vision now cannot function without image processing [5–8]. A variety of medical specialties rely on it for the diagnosis of illness and abnormalities affecting any area of the body. Many disciplines make use of this vocabulary, including healthcare, geography, and multimedia. Radiologists and doctors are medical professionals who analyze medical images. Nevertheless, they employ computer-assisted systems to achieve superior outcomes, as human error and exhaustion are possibilities [9]. Infections can occur as a result of needle-based diagnostic procedures such as biopsies. A second method that is extensively used is X-ray. Cancers of the skin or brain could develop from overuse of this method. PET is the third method utilized for detection. Using a radioactive tracer, this method can produce pictures of the brain. These pictures aid the doctors in diagnosing the illness. Unpleasant pain and illnesses could result, though. Slices are the many perspectives and angles used in a CT scan to take pictures of the brain or any other location. The images you get with this method, though, may be better able to spot a severe illness. One of the most advanced imaging techniques currently used is magnetic resonance imaging (MRI). MRI scans surpass CT scans in terms of quality and accuracy. Nothing ever uses X-rays in MRI. Nobody experienced any discomfort when getting an MRI.

In this research, deep learning techniques along with convolutional neural networks (CNNs) to make MRI scans much more accurate for finding brain tumors have been used. A lot of studies have focused on finding better ways to localize tumors using MRI, improve feature extraction, and classify tumors [10]. Deep learning methods, such as 3D CNNs [11], SVMs [12], and multi-class SVMs, have shown remarkable precision in segmenting and categorizing brain tumors from MRI images. Researchers have found that neural models trained with deep learning may be able to correctly classify MR images by using machine learning algorithms like multilayer perceptron, naïve Bayes, and support vector machines to extract features. MRI and deep learning and machine learning techniques have been looked at as ways to diagnose a number of brain cancers, such as pituitary tumors, gliomas, and meningiomas. Several approaches have demonstrated remarkable efficacy in tumor identification; these include VGG16, SWT, and GCNN [4].

MRI image-based multiclass brain tumor classification presents various critical challenges to automated detection algorithms' accuracy and reliability. Problems include intraclass variability and interclass resemblance. At the same time, distinct kinds may seem visually similar, leading to misclassification rates; tumors of the same type might

range significantly in form, size, and texture. The lack of high-quality, annotated datasets and labeling variations caused by different radiologists' interpretations exacerbate the situation by adding noise to the data. Furthermore, distortions and differences in MRI picture quality induced by various imaging procedures, equipment, and patient placements limit the model's capacity to learn broad features. Furthermore, deep learning models, particularly in clinical contexts, are prone to overfitting on small or unbalanced datasets, affecting generalization to new instances. Another critical challenge in multiclass brain tumor classification is the models' need for interpretability. Doctors often expect clear and logical arguments for forecasts in order to acquire acceptance and credibility in medicine. Furthermore, the technical complexity of merging MRI with other imaging modalities to increase classification accuracy still needs to be addressed. The development and deployment of complex multimodal data fusion algorithms are currently ongoing. These difficulties highlight the critical need for more robust, interpretable, and widely applicable multiclass brain tumor classification approaches.

This research has proposed a hybrid approach that integrates advanced image processing techniques with deep learning to address challenges in multiclass brain tumor classification using MRI scans. The first step is image preprocessing, which uses adaptive noise filtering and Wiener filter-based enhancement techniques to make the MRI quality better by getting rid of artifacts, increasing contrast, and making sure that image clarity is the same for better feature extraction. Next, we employ a stacking ensemble deep learning model to classify various types of brain tumors. Integrating CNNs, trained in separate facets of the data, this technique uses its capabilities. Integrating the forecasts of independent model helps the technique to handle the problems of class imbalance, intraclass variability, and interclass similarity by thus increasing the resilience and accuracy of the classification system. Among all ages, DenseNet121 was chosen because, in addition to the excellent classification performance, it is computationally inexpensive and has relatively few parameters compared to such powerful models as ResNet152 or Inception-ResNet-V2. The increase of connectivity and feature reuse and gradient flow in its connectivity lends it to medical imaging problems where the details must be saved. Moreover, the architectural design of DenseNet121 enhances the quality of visual explanations when using approaches like Grad-CAM, which assists clinicians in interpreting these explanations. Although the improved interpretability and dependability of the findings make the suggested hybrid technique a feasible solution for clinical adoption in brain tumor diagnosis, it guarantees that the model is better generalized and less prone to overfitting. Main contributions of this research are listed below: -.

- The proposed method comprises of fine tuning a pre-trained network (DenseNet121) to harness its unique characteristics, thereby significantly improving classification accuracy and resilience. This method, tailored for clinical use, mitigates the risk of overfitting and enhances generalizability across multiple datasets, effectively addressing issues such as class imbalance and intraclass variability.
- Integrating a modern deep learning model with enhanced image preprocessing in a deep learning approach provides a more accurate and trustworthy way to separate several brain tumor types. This automated diagnostic technology enables radiologists to make intelligent decisions, finally improving patient outcomes by offering a more full and robust detection system.

- Proposed methods use deep learning and image processing methods not just about improving the model's choices, but also about enhancing their interpretability. The preprocessing stage provides more significant visual cues that align with the output of the deep learning model, aiding doctors in understanding the rationale behind predictions and thereby instilling confidence in AI-based solutions in medical diagnostics.

Here is the outline of the article: Brain tumor detection using MRI image analysis is the subject of Sect. 2's literature review on deep learning and machine learning approaches. Section 3 delves deeply into the methodologies and resources utilized, covering everything from datasets and preprocessing techniques to deep learning models, the proposed approach, and the evaluation plan. Section 4 presents the outcomes of the suggested method and various deep learning models, along with an exhaustive description and comparison study of SOTA approaches. In Sect. 5, we draw conclusions, discuss the limitations, and propose new directions for study.

2 Literature review

The diagnosis and treatment of brain tumors have been revolutionized by medical imaging, particularly MRI. AI and deep learning have been looked at in a number of studies as ways to speed up and improve the accuracy of using magnetic resonance imaging (MRI) data to find and classify brain tumors. Based on the research that is currently accessible, this literature review delves into the most recent developments in this field. One significant research direction in machine learning for brain tumors is based on the segmentation and feature extraction, as demonstrated by Khan [13]. Their team came up with a marker-based watershed segmentation, which uses a step-by-step process to improve the contrast and pull-out features from MRI scans. The first step uses gamma contrast stretching to make tumors easier to see. The second step uses a marker-based watershed algorithm for segmentation. The presented algorithm is applied to partition areas that have indistinct borders, which often occur when diagnosing brain tumors. In this framework, features are taken from the shape, texture, and point features. The chi-square-based conditional priority method is then used to pick only the best features. They are then concatenated in series and followed by final classification using the Support Vector Machine (SVM) classifier.

Recent work has further advanced transfer-learning methods in brain tumor imaging. Parida et al. improved the nnU-Net and MedNeXt models for African patients, getting excellent Dice scores of about 0.87 to 0.93 for identifying gliomas [14]. Similarly, Perez et al. introduced HT-CNNs, a new type of model that combines transformers and CNNs, which works well for different types of tumors and performs better than past winners of the BraTS competition [15]. Brain tumor classification advancement using transfer learning techniques has also been provided in the last few years. As an example, Ishaq et al. used EfficientNet to detect glioma subtypes and demonstrated 98.6% accuracy compared to achieved by a DenseNet, which has in reality 14 times more parameters [16]. Similarly, Kaniki et al. explored medical imaging-specific transformer-based vision models and found that they performed better on data acquired in other centers on brain tumors [17].

This study employs shape, texture, and intensity patterns in relation to the ML models for accurate segmentation. Probabilistic Neural Networks (PNNs) were used by Sapra

[18] to separate tumor areas. They showed that PNLs were effective at separating cancerous tissues based on statistical features of pixel intensity. Sharma [19] conducted another study that combines PCA and MLP for tumor classification, achieving an optimal recognition rate of 88.2%. Conventional machine learning models need a lot of manual engineering for their features to be useful, and they use a lot of computing power, so they are not easily scalable. Kumar et al. [20] then used features such as texture and statistical information to classify brain tumors using SVM technology. The classifier provided precisely the localization and segmentation of the tumor in the image matrix, thus enabling early diagnosis. Also, their approach clearly highlighted how SVM is useful in combining feature-based learning with feature segmentation. They first used median filtering and skull masking as preprocessing techniques. Next, texture features were extracted and used to train the SVM. Compared to using single classification methods, the model provided improved outcomes, which supports the use of hybrid models. Upadhyay [21] discussed the use of Random Forest (RF) for the classification of MRI scan images into tumor and non-tumor groups. By changing the images' size, getting rid of noise, and turning them into grayscale, their RF-based system was better at telling the difference between benign and malignant tumors and was more reliable. The use of a confusion matrix and a kappa coefficient provided assurance for the model. In the same way, Naik and Patel [22] used the decision tree classification [109] method, which involved picking out texture features using association rule mining as the feature selection method. The locker decision tree classifier outperforms both Naïve Bayes and the previous classifier's accuracy with a precision level of 100% and an accuracy of 96%. The study also highlighted how decision tree algorithms can efficiently work with large MRI datasets.

Not long ago [23], Saeedi wrote about a 2D CNN and a convolutional auto-encoder that worked well on a set of data that had gliomas, meningiomas, pituitary gland tumors, and normal brain MRIs with no problems. The CNN-based method undertaken offered a training accuracy of 96.47% with the recall values of 95% and DCA AUC approximately equal to 1. A study by Saeedi et al. backed up this claim by looking at CNNs and other machine learning models, like KNN-ML and MLP. They found that CNN was the best for classifying brain tumors. Radiologists can easily integrate this proposed model into clinical systems for real-time diagnosis due to its efficiency. CNN has yet another advancement in the domain of DL-based brain tumor detection, which was another breakthrough. CNNs are especially suited for image-based analyses because they can identify the spatial pyramid of data. Sadad et al. made a CNN-based model with the Unet architecture and ResNet50 as its main part in late 2021 to find different kinds of brain tumors, like gliomas, meningiomas, and pituitary tumors [24]. This work showed big improvements over the usual CNN by making it easier to extract features by adding more data and preprocessing it. This resulted in an IoU of 0.9504, making the model efficient in multi-class classification. The detection of brain tumors has also utilized several CNN architectures, such as DenseNet201, MobileNet V2, and NASNet, each contributing additional features to the system. In the work of Sadad et al., NASNet [25], which was used to study certain objects, got a 99.6% success rate, which is better than other CNN models. State-of-the-art pre-training techniques, such as transfer learning and reinforcement learning, rely on transfer weights to design and build these models. As a result, they require a relatively limited dataset for training while maintaining high accuracy.

Saba [26] employs GrabCut for segmentation and VGG-19 for feature extraction in their approach. We combine shape and texture descriptors with deep features from VGG-19 to create a fused feature set. We do this because each location's handcrafted features are unique. Entropy is the most effective method for feature fusion, as it improves the accuracy and speed of the classifiers' classifications. We then train classifiers on different brain tumor datasets, including BRATS 2015, 2016, 2017, and 2018, using the combined set of features. The DSC reached a high point of 1.00, showing that the hybrid concept is a very effective way to improve diagnostic accuracy. Amin [27] used an LSTM-based model for promoting the classification of the brain MRI scans. The model was able to find temporal features that are important for modeling how tumors change over time after processing these sequences with LSTM. The proposed model did very well when tested on different BRATS datasets; for example, it had a Dice Similarity Coefficient of one for the BRATS 2012 synthetic database. These results clearly justify the use of LSTM in complex diagnostic processes where data analysis is important. Sharif et al. worked [28] on feature selection using the PSO algorithm, in which knocking out ineffective features improved the classification accuracy. That's how the Dice coefficient for core tumors on the BRATS 2017 data set got as high as 93.7% when this optimization strategy was used with CNN and saliency-based segmentation techniques. This suggests that using advanced optimization techniques along with deep learning-based models leads to better diagnostic results. Nazir et al. [29] studied CNN-based models from 2015 to 2020 and found that using DNNs to accurately identify brain tumors worked. The study introduced different architectures, such as transfer learning and fine-tuning, to tackle challenges affecting deep learning models. Additionally, Nazir said that this would continue to be the main goal in order to improve intertwinement or make models easier to understand and explain, which is becoming more and more useful in clinical practice. Using ResNet and GoogleNet, Harem Kibriya et al. introduced a multiclass approach to brain tumor classification. They looked at three distinct tumor types and discovered that ResNet had a 98.0% accuracy rate and GoogleNet 97.6%. In a similar vein, P. Gokila Brindha et al. utilized ANN and CNN to create a method for binary classification of brain cancers. According to their results, CNN got 89% accuracy while ANN got 65.21% [30]. Using average pooling convolutional neural networks, Jagadeesh Kakarla et al. demonstrated a three-class classification system that had a success rate of 97.42% [31]. Using CNN, Díaz-Pernas et al. achieved an accuracy rate of 97.3% in classifying and segmenting three different kinds of brain tumors. Using convolutional neural networks (CNN) and support vector machines (SVM), Angona Biswas et al. came up with a 96% accurate model for classifying glioma, meningioma, and pituitary tumors [24]. Using a small dataset, Sarah Zuhair Kurdi et al. achieved 98% accuracy in binary brain tumor classification using CNN. Prasun Chandra Tripathi et al. developed a model based on deep residual networks that accurately detected two kinds of gliomas 95.67% of the time [32]. Last but not least, S. Deepak et al. created a CNN-SVM hybrid model that correctly categorized three types of brain tumors 95% of the time and did well in other tests [33].

Zahid et al. investigated classification techniques to find a way to make the ResNet101 model better by combining MRI sequences from different imaging sources. Scientists used particle swarm optimization and differential evolution to find the most important parts of the model, even though it created useless features that changed both how

accurate it was and how much processing power it needed. We further limited the options using Principal Component Analysis (PCA) before aggregating less significant features. Reaching a 96.7% accuracy, a medium neural network handled the last categorization. Although it turned out successful, the fusion process increased the computational time required [34]. In the same way, Muhammad et al. came up with a new way to subtype brain tumors by combining particle swarm optimization with deep CNN models. Their method dramatically reduced the number of false positives when distinguishing between benign and malignant tumors. Their segmentation accuracy was 98.11%, and their classification accuracy was 98.25% on the BraTS 2018 dataset, for a total success rate of 98.9%. By reducing the need for human input and establishing the PSO threshold value automatically, further optimization could enhance productivity [35].

Desouza et al. used radiomic features along with neural and ensemble learning models to sort MRI sequences of different types (like T1, T2, and FLAIR) into groups that correspond to benign gliomas, meningiomas, or pituitary tumors. They discovered that XGBoost, or Extreme Gradient Boosting, outperformed the deep neural network. Using K-fold cross-validation, its accuracy was 88.51%, while this method achieved 87.09%. Their investigation was limited to specific MRI sequences because it did not include the T1CE MRI modality [36]. Similarly, Gupta et al. developed a rapid method for classifying brain cancers using FLAIR pictures from 200 patients. We used Discrete Wavelet Transform (DWT) and Principal Component Analysis (PCA) to get the features and pick the twelve largest slices that would be used as classification features. K-fold cross-validation was used to test different classifiers, such as Random Forest, Support Vector Machine (SVM), K-nearest Neighbor (k-NN), and Classification and Regression Tree (CART). The best classifier, according to SVM, achieved an accuracy of 88%. However, the approach has difficulty distinguishing malignancies near the scalp from other, brighter objects in the FLAIR series [37].

In their study, Rajasree et al. suggested a way to look at magnetic resonance imaging (MRI) images that uses both a multiscale multimodal convolutional neural network (MSMCNN) and a long short-term memory (LSTM) technique for deep learning-based semantic segmentation. Scientists can assign predefined classes to individual pixels in an image using this method. The model is built on a multiscale U-Net deep convolutional network, which handles multimodal convolutions with three different pixel levels. For tumor classification, the LSTM network combines these three convolution routes. Although this method needs clinical validation and more datasets for practical training, it attained a 96% accuracy when applied to the BraTS 15 dataset [38]. Anantharajan et al. used MRI images and subjected to preprocessing using an Adaptive Contrast Enhancement Algorithm (ACEA) followed by a median filtering technique to improve image quality. The enhanced images are then segmented using a Fuzzy C-Means clustering approach. Key texture features such as energy, mean, entropy, and contrast are extracted through the Gray-Level Co-occurrence Matrix (GLCM). Finally, an Ensemble Deep Neural Support Vector Machine (EDN-SVM) classifier is employed to accurately classify the abnormal tissue regions [39]. Guo et al. did their own study to improve the accuracy of glioma subtype classification by using a mix of T1, T2, T1ce, and Flair MRI techniques. They used a fusion-based network's segmented features to feed a DenseNet model. To draw conclusions, we used a decision-fusion method with a linear weighted

module to put together the probabilities that the pre-trained models said would happen. Their strategy was able to reach an 87% success rate [40].

3 Materials and methods

Central to the suggested approach for brain tumor classification is the DenseNet121 deep learning algorithm. The three primary levels of the proposed transfer-learning-based system for brain tumor identification are data collecting, preprocessing, and application. The first layer of data acquisition is responsible for collecting raw data. The preprocessing layer is accountable for managing, moving, and normalizing this raw input. Next, the DenseNet121 model is used in the third tier's prediction layer with the processed and consistent data. Variables serve as input and outputs to the data acquisition layer. The last layer, performance evaluation, determines the accuracy of the system by calculating precision and miss rates. We employed 5-fold cross-validation to assess the model's performance. The results are reported as mean accuracy $\pm$ standard deviation along with 95% confidence intervals to ensure statistical reliability and reproducibility. Based on the extensive analysis performed throughout the layer, conclusions are generated on the development of brain tumors in the concluding phase.

3.1 Image acquisition

Figure 1 represents the proposed scheme for the classification of brain tumors. The IoMT sensor layer is affixed with the input sensors for collecting relevant data from different devices like system hardware, laptops, cameras, and smart devices like direct from MRI imaging machines [28]. Then, it is sent to a preprocessing layer, which is responsible for reducing the noise that occurs due to wireless communication. This layer supports radiologists and neurologists in making appropriate diagnoses and treatment decisions.

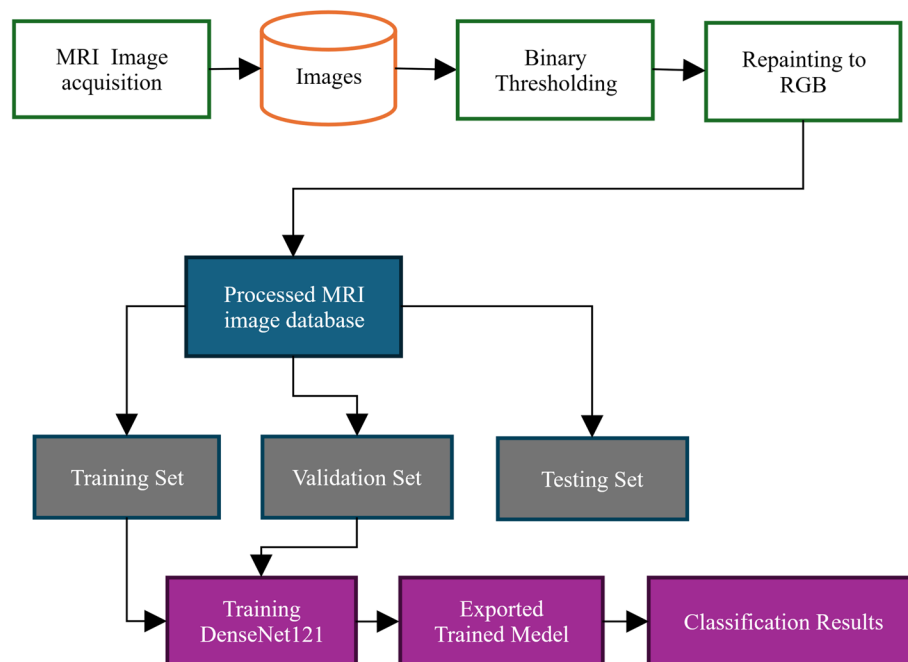


Fig. 1 Proposed Multiclass brain tumor classification framework

3.2 Datasets

The Brain Tumor MRI dataset, with 7023 brain MRI images, serves as the standard for applying the suggested approach. This dataset, sourced from Figshare, SARTAJ, and Br35H datasets [29], has been selected for its comprehensive coverage of different brain tumor types. The image categories include Glioma, Meningioma, No Tumor, and Pituitary. Images classified as No Tumor were extracted from the Br35H dataset. Due to labeling issues in the SARTAJ dataset's Glioma category, images from Figshare were utilized instead. The MRI dataset of the present study has numerous modalities (T1- weighted, T2- weighted, FLAIR, DWI, among others). T1-weight images give well-anatomical structure with superb white and gray matters contrast. T2 images emphasize regions with fluid tissue and are helpful in the detection of edema and cysts. FLAIR sequences diminish signals of the cerebrospinal fluid thereby enhancing the visibility of the lesions that are close to ventricles. The abnormality in diffusion is detected with the DWI sequences, the earliest indication of a pathological difficulty like ischemia or cellular density in malignancies [41]. The dataset is available for download at <https://www.kaggle.com/datasets/masoudnickparvar/braintumor-mri-dataset>. Sample images from the dataset are depicted in Fig. 2. Although the Kaggle dataset integrates MRI scans from multiple publicly available sources (Br35H, Figshare, and SARTAJ), it may not capture the full variability found in clinical imaging practices. As such, the results may not fully generalize across different institutions or imaging protocols.

We have set aside 80% of the data for training and 20% for split testing. Previous research and real-world data informed the selection of this popular ratio, which strikes a good compromise between building a robust model and retaining a substantial amount of data for performance evaluation. This split makes sense since outstanding prediction accuracy depends on the model learning from a complete and representative collection of training samples. By using 80% of the data for training, the model gains access to a broad spectrum of samples, thereby enabling its identification of significant trends and features intrinsic to the data. An independent dataset for assessing the generalizing capacity and performance of the model on unprocessed data is the 20% set aside for testing. Table 1 presents a complete study of the distribution of the dataset into the training

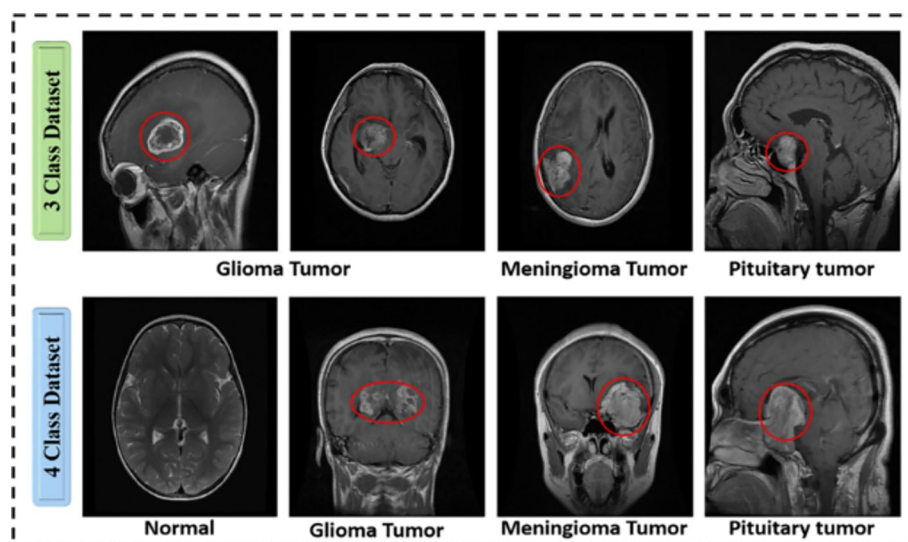
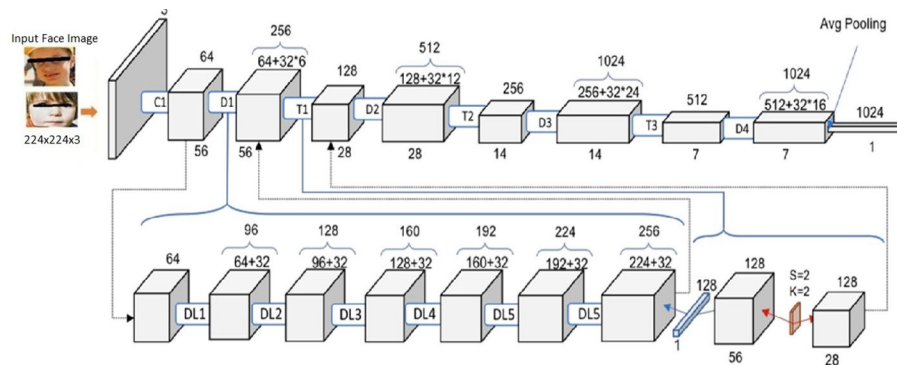


Fig. 2 Sample Images from Multiclass merged brain tumor dataset

Table 1 Distribution of images of datasets

Dataset	Glioma	Meningioma	Pituitary	Normal	Total
Training	1321	1339	1457	1595	5712
Testing	300	306	300	405	1311
Total	1621	1645	1757	2000	7023


Fig. 3 DenseNet-121 architecture

and testing sets coupled with the number of images or samples allocated to each other. This table clearly summarizes the data split and guarantees openness in the selected method for model evaluation.

3.3 Preprocessing

Data preprocessing plays a significant role in preparing MRI images for brain tumor identification. The existence of noise is one of the main difficulties since it can compromise correct diagnosis. The work incorporates a preprocessing module to handle this. Techniques of image enhancement are used to raise visual clarity and enable easier analysis. All MRI pictures are first scaled to $224 \times 224 \times 3$; this guarantees constant input dimensions and maximizes computing performance. Labeling helps improve model performance. We applied class-specific data augmentation to address the class imbalance observed in the dataset (e.g., 2,000 “normal” vs. 1,621 glioma images). For classes with fewer images, like gliomas and pituitary tumors, we used techniques like random rotations ($\pm 15^\circ$), flipping the images sideways and up and down, zooming in, and changing the contrast to increase the number of training images and help the model learn better. We use image augmentation to increase data diversity, which in turn enhances the accuracy of the model. Among the other changes used are a 15-degree rotation range, a 0.1 width and height shift, a 0.1 shear range, a 0.1 zoom range, horizontal flipping enabled, and a “Nearest” fill mode set. Eliminating noise from MRI pictures is still difficult, though, since many denoising methods cause blurring effects and other artifacts [42]. This work reduces noise by means of median and CLAHE filters; therefore, it maintains important image features.

3.4 Proposed DenseNet121 architecture

Gao Huang unveiled DenseNet121, a deep-learning model showing in Fig. 3. There are 121 layers total, structured into dense blocks of each constant number of layers. The standard RGB image size for feeding data into a neural network is 224 by 224 pixels. The model’s transition layers apply batch normalization and down-sample the feature

map. The ReLU activation functions are applied to the model after each convolutional layer. With the use of auxiliary layers like SOFTMAX output and global average pooling, DenseNet121 can manage picture recognition jobs with little vanishing gradient issues efficiently. We may observe the dense block in action here:

$$X_l = H_l([X_0, X_1, \dots, X_{l-1}]) \quad (1)$$

Here, the output of the i th layer in the dense block is denoted by X_l . H_l denotes the composition of operations performed at layer l . The notation $[X_0, X_1, \dots, X_{l-1}]$ represents the concatenation of the feature maps from each dense block layer that came before it (see Fig. 3).

This study involved fine-tuning pre-trained models by incorporating updated final layers while keeping the other layers frozen. Subsequently, the researchers employed transfer learning to categorize images photos for brain tumor identification. This enabled the underlying model to prioritize the acquisition of new custom layers for training fresh data rather than erasing its previous knowledge. The additional layers substitute for the fully linked layer of the pre-trained models. The model employs a pre-trained base to incorporate transfer learning. Before inputting the $224 \times 224 \times 3$ pictures into pre-trained models for automatic feature extraction, we subjected them to data preprocessing and augmentation techniques. Figure 4 displays the recommended final layers for pre-trained models, which include a batch normalization layer, a flattening layer, two dense layers with ReLU activation, a dropout layer, and a SOFTMAX classifier. The initial layer employs batch normalization to normalize the input data. Next, the feature maps with several dimensions are flattened into a one-dimensional array. To promote non-linearity in the model, the dense layer below employs a rectified linear unit (RELU) activation function of 512 units. A supplementary batch normalization layer is then incorporated to stabilize and accelerate the training process. The complexity of the model increases further with the inclusion of the dense layer, which also consists of 512 units and has a RELU activation function. A dropout layer with a dropout rate of 0.2 is employed during training to deactivate a portion of neurons, hence preventing overfitting randomly. The classification process is finalized by a dense layer consisting of two neurons, each representing one of the output classifications.

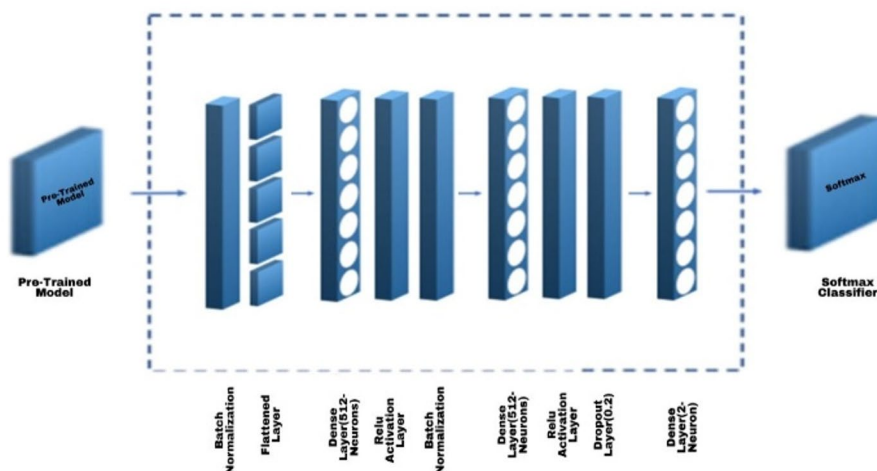


Fig. 4 Proposed layers for transfer learning

3.5 Experimental setup

First, we prepare and enhance the dataset. Second, we use hyperparameters to fine-tune the deep convolutional neural network (CNN) models. Finally, we evaluate the models using appropriate performance metrics. This is where the Kaggle repository's dataset of MRI brain tumor comes into play. Experiments were conducted on different Python libraries and hardware devices.

3.5.1 Hyper parameters

We exploited all the hyperparameters for the deep learning models, as compiled in Table 2, during training. These hyperparameters were deliberately chosen to improve performance. Using a learning rate of 1e-4 will help to guarantee slow model convergence with our dataset and enable fine-tuning. Empirical research validated this value and showed it to be rather successful. Using a batch size of 32 allowed us to also maximize memory use and training speed depending on our hardware setup. This set aimed to balance accuracy and efficiency, considering GPU resources and memory overflow prevention. Finally, we decided on SOFTMAX as the classifier. The SoftMax classifier layer mostly serves to decide the final image categorization and associated category. Applying the SoftMax function to the output creates a probability distribution spanning all classes. The SoftMax function generates the likelihood of an image belonging to each category using the following equations.

$$S(Z_i) = \frac{\exp(Z_i)}{\sum_{j=1}^k \exp(Z_j)} \quad (2)$$

In the preceding equation, Z represents the input vector of length k , whereas $\exp(Z_i)$ denotes the exponential of the i -th element, and the total of the exponential values over all elements in the input vector Z is given by $\sum_j \exp(Z_j)$. Finally, $S(Z_i)$ produces a probability distribution of images over the k classes. We employed categorical cross-entropy to calculate the average difference between expected and predicted values. Equation 3, which shows the loss measurement for the binary classification, uses the binary values 0 or 1 whereas y_{ij} to represent binary values and the probability p_{ij} .

$$\text{CE Loss} = -\frac{1}{N} \sum \sum y_{ij} \cdot \log(p_{ij})_{j=1}^{NC^N} \quad (3)$$

Choosing the Adam RMS Prop method as an optimizer allowed us to decrease loss as much as possible during training. With an adaptive gradient descent function, this optimization method speeds up the process of weights convergent to local minimum. We trained and tested over a 50-epoch period to increase the robustness of the pre-trained models and minimize problems with underfitting and overfitting.

Table 2 Hyper parameters used to train proposed model

Parameter	Value	Parameter	Value
Image Size	224 × 224	Learning rate	1e-4
Activation function	RELU	Optimizer	RMSprop
Output classification layer	SOFTMAX	Loss-function	Categorical-Cross-Entropy
Dense Layer	512	Batch -size	32
Dropout value	0.2	Epochs	50

3.6 Evaluation measures

In this study, performance metrics like ROC, F1-score, recall, accuracy, and precision are used to assess the pre-trained models. One popular numerical method for evaluating a classification model's performance is the confusion matrix. They offer a systematic look at the model's accurate and incorrect predictions during testing. When testing a trained model's predictive power on a validation dataset, the confusion matrix (CM) is a typical tool to employ [43–47]. The rows and columns of the matrix represent the actual categories of cancers, including gliomas, meningiomas, pituitary tumors, and non-tumors. The following are mathematical formulas for ROC, F1-score, recall, accuracy, and precision:

A key metric used to measure the overall effectiveness of a classification model is its accuracy. This is calculated by dividing the total number of correct predictions, both positive and negative, by the total number of instances in the dataset. The accuracy can be expressed using the following formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + TN + FP} * 100 \quad (4)$$

Precision is a measure of how accurately a categorization model predicts favorable outcomes. The sum of True Positives and False Positives yields the total count. Here is the formula for precision:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

The recall of a classification model is a measure of its performance that measures how accurately it identifies all relevant instances of a positive class. The words sensitivity and true positive rate are further synonyms for recall. It is determined by dividing the total number of True Positives and False Negatives by the number of True Positives only. Here is the recall formula:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

One functional performance statistic is the F1 score, also called the F1 measure, which takes precision and recall into account in a confusion matrix. When weighing the relative importance of recall and precision, it becomes beneficial. The following equation is used to get the F1 score:

$$\text{F1 - Score} = \frac{2TP}{2TP + FP + FN} * 100 \quad (7)$$

The Receiver Operating Characteristic (ROC) curve is a graphical representation that illustrates a classification model's performance across multiple thresholds. It is created by plotting the true positive rate against the false positive rate at each threshold level.

4 Results and discussions

The result section describes the results detail obtained by proposed DenseNet121 model and comparison with different deep learning models based on transfer learning for classification of four major classes of brain tumor through MRI images. The following steps are included in this section:

- Performance analysis of Proposed Model.
- Comparison of proposed with other deep learning models.
- Comparative analysis of proposed and state-of-the-art methods.

4.1 Performance analysis of proposed method

In Fig. 5 the graphs above show how well a DenseNet121 model trained and tested over 50 times to be able to classify brain tumors into multiple groups. These graphs represent accuracy and loss curves, which are essential for assessing the model's effectiveness during training. Looking at the left training and validation accuracy graph, the model first shows a fast rise in both during the first epoch. This rapid increase shows DenseNet121 is learning from the dataset and modifying its parameters to raise classification accuracy. The model keeps improving its knowledge of the data as training goes; training accuracy steadily approaches 100% and stabilizes. Comparably, validation accuracy rises rapidly at first but subsequently levels out and somewhat falls below training accuracy. This trend shows that the model performs well, but the slight accuracy decline over validation may indicate that the set is too complex to capture or fit too well.

The right-hand side of the graph shows the training and validation loss, which further substantiates the model's learning efficiency. Both loss values decrease sharply early in training, suggesting the model quickly improves prediction accuracy and decreases error rates. The training and validation dataset losses level off at low levels after the first quick drop. This suggests that the model has probably found its best level of performance. Due to the lack of discernible difference between the training and validation loss curves, the model is free from severe overfitting. DenseNet121 architecture excels in generalizing models. The model learned brain tumor images well and stayed accurate on new data because it had low and stable loss values and a small difference in accuracy between training and validation. Still, the slight variation in the training and validation accuracies suggests some likely overfitting, a typical problem in deep learning projects using demanding datasets like medical imaging. Strategies include data augmentation, drop-out layers, or other regularization methods that could even help the model to generalize further.

According to our results, the DenseNet121 model appears to be well-suited to the multiclass brain tumor classification task. Low-loss levels and high accuracy indicate robust learning and accurate predictions. Therefore, in order to further improve the model and reduce any overfitting tendencies, future research should focus on either expanding the

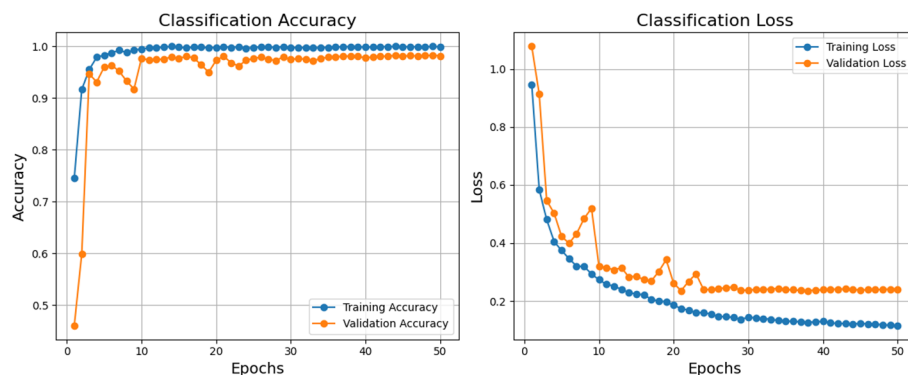


Fig. 5 Training and validation accuracy and loss curves of the proposed model, illustrating its performance and convergence behavior during the learning process

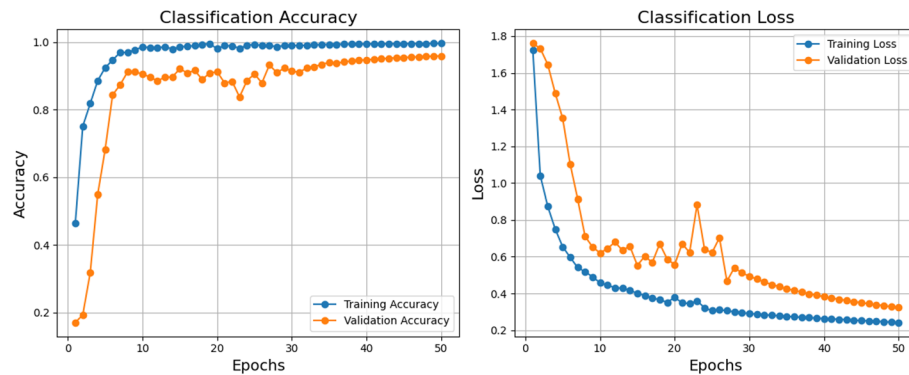


Fig. 6 Training and validation accuracy and loss curves for DenseNet121 without modification, demonstrating the model's learning performance over epochs

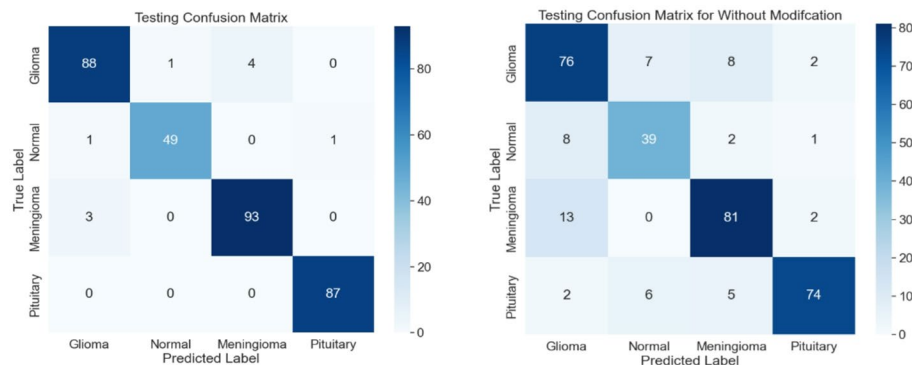


Fig. 7 Confusion Matrix Comparison on Test Set: (Left) Proposed DenseNet121 Model, (Right) Original Pretrained DenseNet121 Model

dataset or enhancing the model's architecture. Examining cases when the model underperforms or misclassifies could potentially provide valuable insights that direct improvements to raise its performance in pragmatic, real-world clinical environments.

Without any changes, Fig. 6 shows over 50 epochs, classification accuracy (left), and classification loss (right) of the DenseNet121 model. The training accuracy rises steadily and rapidly plateaus close to 1.0, so indicates the model's capacity for efficient training data fit. Though it shows some fluctuations, indicating minor instability in generalization across validation data, the validation accuracy also improves dramatically during the first epochs and stabilizes at about 0.85. The training loss steadily lowers on the classification loss graph and reaches a low point. The validation loss likewise decreases dramatically in the early epochs at the same time, but it varies somewhat following epoch 10. Beyond this point, this difference between training and validation loss suggests potential overfitting when the model struggles to maintain constant performance on unseen validation data while becoming highly optimal for the training data. Though further optimization, e.g., regularization approaches or data augmentation, may help increase validation stability and generalization, the overall trend implies solid learning progress and satisfactory performance despite these variations.

Specifically, glioma, meningioma, pituitary tumors, and detecting regular brain images; the confusion matrix displayed in Fig. 7 indicates the usefulness of a DenseNet121-based model in finding numerous forms of brain tumors. The matrix reveals that the model

performs somewhat well over the four classes, with remarkable accuracy in determining every category. In 88 cases, for example, gliomas were accurately categorized; there were only a few misclassifications—one instance was projected as usual and 4 as meningioma, therefore indicating minor confusion with other tumor types. Regular brain scans were similarly primarily accurate, with only one misclassification as glioma to highlight the model's strength in differentiating healthy tissue. The model proved its remarkable capacity to differentiate between tumor types since it attained a high degree of accuracy in the case of meningioma tumors and, in just three instances, misclassified as gliomas. Moreover, the perfect classification of pituitary tumors—no misclassification—showcased the effectiveness of the algorithm in recognizing this type. The significant number of accurate predictions across the board indicates that DenseNet121 efficiently detects complex patterns and characteristics inside MRI images, enabling precise classification.

The DenseNet121-based model for multiclass brain tumor identification has Receiver Operating Characteristic (ROC) curves, which can be seen in Fig. 8. You can see how well the model distinguishes between glioma, normal, meningioma, and pituitary tumors by plotting their curves. The AUC values are exceptionally high across the board. These numbers provide a good indicator of the model's ability to distinguish between categories. The glioma class achieved excellent results in reducing false positives and identifying glioma patients, with an AUC of 0.965. The average class outperformed the model in detecting cancer-free healthy brain pictures, achieving an AUC of 0.979. Consistent categorization of meningiomas was demonstrated by a low false positive rate of 0.976. Pituitary tumors were a standout category, with an AUC of 0.998. Inferring near-perfect categorization with little overlap between categories, the model does a fantastic job of distinguishing pituitary tumors from other types. When it comes to identifying and classifying different types of brain tumors, the DenseNet121 model typically has excellent ROC curves. The strong AUC ratings suggest that model learning might successfully generalize the test data, especially for pituitary tumors. These findings demonstrate the capability of DenseNet121 to detect brain cancers, paving the way for prompt diagnosis and treatment planning. On the other hand, small differences in AUC values between classifications, especially glioma, show that the model could use some small changes.

To ensure the reliability and the repeatability of our results, we conducted 5-fold cross-validation on the training dataset. The key metrics applied in assessing the performance of the model comprised accuracy, precision, recall, and F1-score. We present each measure as the mean $\pm$ standard deviation across the five folds. Also, we calculated the 96%

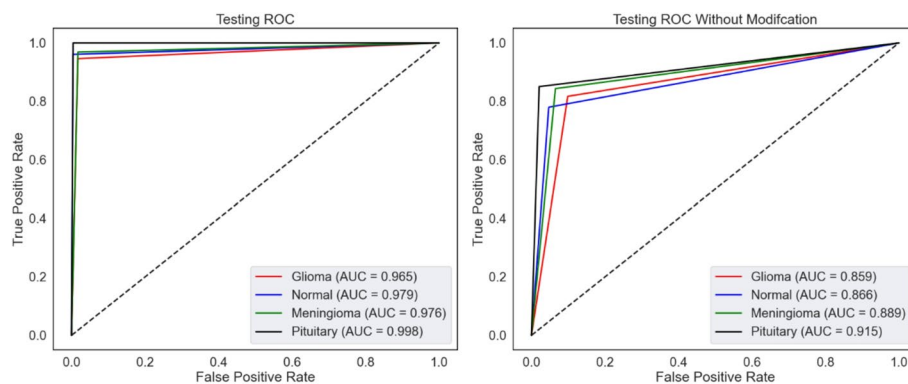


Fig. 8 ROC Curve on Test Set: (Left) Proposed DenseNet121 Model, (Right) Original Pretrained DenseNet121 Model

Table 3 Average performance across the cross-validation folds

Metric	Mean ± Std Dev	96% Confidence Interval
Accuracy	97.3% ± 0.8%	[96.1%, 98.5%]
Precision	97.1% ± 0.9%	[95.8%, 98.4%]
Recall	97.6% ± 0.7%	[96.5%, 98.7%]
F1-Score	97.3% ± 0.8%	[96.1%, 98.5%]

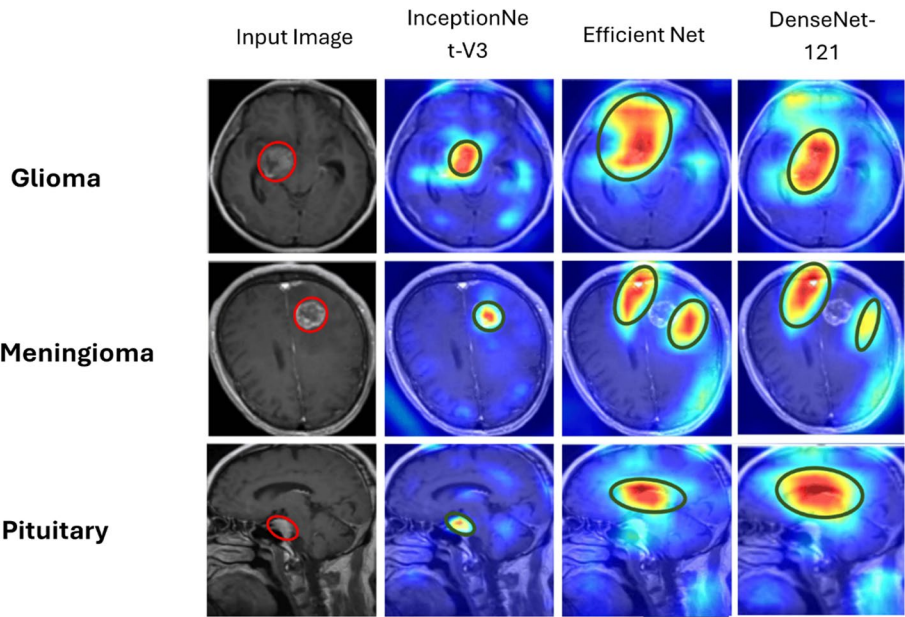


Fig. 9 Grad-CAM visualizations: (Left) Correct classifications with focused tumor activation; (Right) Misclassifications with misleading region focus

confidence interval to enable us to measure the statistical reliability of the results. These similar outcomes on folds attest that the model is highly generalizable and causes little chance of overfitting. The statistical validity of our results is also enhanced because of the inclusion of confidence intervals (see Table 3).

4.2 Classification and error analysis

To better understand the clinical results, we aimed to visualize the areas of the MRI images that most influenced the model predictions; for this purpose, we used Gradient-weighted Class Activation Mapping (Grad-CAM). A key step toward clinical integration is improving interpretability. Incorporating Grad-CAM has allowed us to provide visual explanations of model predictions. These tools offer transparency to clinicians by showing tumor-specific activations. In future work, we aim to involve radiologists directly in evaluating these explanations and co-developing clinician-friendly diagnostic interfaces. Figure 9 represents the heat maps of correctly and incorrectly represented tumor cases of the four tumor classes. The outlined areas are quite close to the positions of the tumor, and this gives visual confirmation to the output of the model. These types of interpretability tools allow clinicians to check their expectations and identify possible misclassification. Figure 9 shows activation maps created with Grad-CAM for three types of brain tumors: glioma, meningioma, and pituitary, comparing how well InceptionNet-V3, EfficientNet, and DenseNet-121 understand them. Column 1 displays the original

MRI scan, highlighting the tumor areas in red. The following columns are heat maps of areas that every model implemented to classify. In the correctly labeled examples (on the left side of each model's output), the focus is strongly on the actual tumor areas, matching the red highlights. However, on the right side (where the models made mistakes), the attention maps incorrectly focus on unimportant areas, with EfficientNet and DenseNet-121 being more likely to overfit or confuse patterns due to the background designs. This visual demonstration emphasizes the need to develop interpretability for specific certifiable clinical implementations.

The DenseNet121-based multiclass brain tumor classification model was able to correctly group the four types of tumor shown in Table 4: glioma, normal, meningioma, and pituitary tumor. Across all classes, the model achieved outstanding reliability and longevity while also achieving recall, accuracy, and F1-score. The model's accuracy in detecting gliomas was 95.70%, and its recall was 94.60%. As a result, its overall accuracy was 96.90%, and its F1 score was 95.10%. These results demonstrate the algorithm's impressive glioma-identifying capabilities, with only rare misclassifications, but there is still a little trade-off between recall and accuracy. With a recall of 96.10% and a precision of 98.00%, the usual class achieved an F1 score of 97.00%, which was even better than before. This shows that the model can effectively differentiate between normal brain scans and diseased ones, even though it occasionally produces false positives.

The meningioma classification also achieved strong results, with a recall of 96.00% and a precision of 95.00%, resulting in an F1 score of 96.40%. This provides evidence that the model can reliably identify meningioma cases on a regular basis. The model did an outstanding job in pituitary tumors, with a recall of 100%, a precision of 98.90%, and an F1-score of 99.40%. Given the model's exceptional accuracy in detecting pituitary tumors, this nearly perfect classification implies that it leaves essentially no room for false negatives. Overall, the DenseNet121 model has demonstrated balanced performance and remarkable accuracy for all forms of cancer. Even though the precision and recall scores are always good, more work may need to be done to improve detection in groups like glioblastoma where small differences in recall and precision mean that the results are sometimes wrong. Future research should concentrate on improving the model's ability to distinguish between comparable tumor types by incorporating additional data or assembling other models. As a result, the model's classification capabilities will be enhanced.

4.3 Performance comparison with other deep learning models

Table 5 shows that the suggested DenseNet121 model does a better job of diagnosing different types of brain tumors in the same testing environment than a number of other

Table 4 Class wise testing results of proposed model on test dataset

Model	Class	Precision	Recall	F1-Score	Accuracy
Proposed DenseNet121	Glioma	95.70	94.60	95.10	96.90
	Normal	98.00	96.10	97.00	
	Meningioma	95.90	96.90	96.40	
	Pituitary	98.90	100	99.40	
Pretrained DenseNet121	Glioma	76.80	81.70	79.20	82.80
	Normal	75.00	78.00	76.50	
	Meningioma	84.40	84.40	84.40	
	Pituitary	93.70	85.10	89.20	

deep learning architectures. DenseNet121 ranks first in all metrics, including accuracy (96.90%), precision (97.20%), recall (97.15%), and F1-score (97.10%). This means the model is excellent in detecting brain tumors, according to its well-rounded and robust performance overall assessment criteria. Despite ResNet-50's impressive performance, DenseNet121 outperforms it in terms of recall and precision. Not as well as competing designs, DenseNet121 only manages a 94.10% accuracy and a 93.80% F1 score. Since VGG-16's 92.90% accuracy and 92.50% F1 score suggest it may not adequately capture the many aspects of brain tumor classification, it needs more time to catch up. Instead of important precision and recall metrics, InceptionV3 and EfficientNet both achieve competitive accuracy rates of 94.81% and 95.60%, respectively. In terms of achieving a balance between accuracy and the F1 score, DenseNet121 still outperforms the recommended model. Given its simplistic design and limited capacity to interpret complex feature hierarchies in medical images, it is not surprising that AlexNet's previous design demonstrates the lowest performance, with an accuracy of 91.00% and an F1 score of 90.50%. While DenseNet121 has achieved unprecedented levels of accuracy, Xception functions in a manner comparable to EfficientNet.

With its deeper design and practical feature reuse, DenseNet121 obviously shows better performance in identifying brain cancers across all four classes. Comparatively to other well-known deep learning models, the results show DenseNet121's strength in extracting complex patterns in brain MRI data, hence improving classification accuracy and lowering misclassification rates. These results imply that DenseNet121 is a viable model for valuable applications in medical imaging, especially in the automated diagnosis of brain tumours, providing high precision and dependability that are very important in healthcare diagnostics.

4.4 Performance comparison with existing approaches

We compare the suggested strategy to current methods for brain tumor detection and classification. Compared to other methods, this one stands out for how well it works and how important it is. The table demonstrates the high accuracy and excellent performance of the proposed DenseNet121 multi-classification model. The proposed DenseNet121-based multiclass brain cancer detection ranks among the best performers when compared to present methods since the performance results shown in Table 6 demonstrate a competitive accuracy of 96.90%. The proposed model beats many current techniques using several deep learning architectures. For instance, the method in [38], 2022, which combined various pre-trained CNN models—including ResNet50 and many variants of EfficientNet—achieved an accuracy of only 87.67%. Particularly in acquiring relevant characteristics for brain tumor categorization, DenseNet121 shows a substantial enhancement. Comparably, by merging public CNN architectures like ResNet50,

Table 5 Comparison with other deep learning models in same settings

Model	Precision	Recall	F1 Score	Accuracy
DenseNet121 (Proposed)	97.20	97.15	97.00	96.90
ResNet-50	94.50	94.00	93.80	94.10
VGG-16	93.00	92.80	92.50	92.90
InceptionV3	95.00	94.70	94.60	94.80
Efficient Net	96.00	95.50	95.40	95.60
Alex Net	91.00	90.70	90.50	91.00
Xception	95.50	95.00	94.90	95.10

Table 6 Comparison with existing methods

Reference, Year	Method	Accuracy
[48], 2022	Pretrained EfficientNetV2B1.	87.67%
[49], 2023	Pretrained ResNet50	95.12%
[42], 2021	IoT based CNN-MRFO Model	97.57%
[50], 2021	IoMT-Deep learning neural network Model	90%
[51], 2024	Feature-enhanced deep learning technique	95.1%
[52], 2024	Deep CNN	92%
[53], 2024	ARM-Net	96.64%
[54], 2025	DieT Transformer model with PCA-ADE integration	96.02%
[55], 2025	Image fusion based Explainable AI	94.28%
[56], 2025	Optimized attention-based lightweight CNN using particle swarm optimization	95.04%
Proposed Method	Dense Net121 multiclass detection of brain. tumor	96.90%

Inception V3, Xception, MobileNetV2, and EfficientNetB0, the model given in [39], 2023, acquired a greater accuracy of 95.12%. This approach shows good performance, but it still has to catch up to the accuracy of 96.90% of the DenseNet121 model, suggesting that the recommended model’s deeper feature extraction powers a more accurate classification of complex MRI data. With an accuracy of 97.57%, the DenseNet121 model falls somewhat behind the IoT-based CNN-MRFO model from [37]. Still, the slight difference in the dataset used in these tests and several model optimization strategies help explain it. Conversely, the IoMT-Deep Learning Neural Network Model from [40], achieved just 90% accuracy, implying that more traditional deep learning techniques potentially lack the endurance afforded by DenseNet121’s architecture.

Methods for extracting meaningful information from clinical sweeps have been developed and implemented in this study. A vital task is the analysis of brain tumors utilizing medical imaging. The detection process is further complicated due to the many difficulties linked to its diagnosis. Radiologists and neurologists primarily perform manual diagnoses. A lot of people are interested in neural networks. According to the results, radiologists and neurologists can alleviate the burden of the massive screening condition by employing the suggested DenseNet121 for multiclass brain tumor detection framework for brain tumor malady analysis. The offered framework is reliable and can be successfully implemented in therapeutic divisions to provide appealing results. This paper presents a system that is both feasible and aesthetically pleasing, which can aid medical professionals by simplifying their work and displaying the likelihood of specific outcomes. Better, more accurate results can be obtained by using DenseNet121. Even though we made all images as consistent as possible, we understand that in the case of different sources, such as different resolutions and contrast in datasets—in datasets of Figshare, SARTA], and Br35H, these factors may cause small bias contributors. Future prospectuses can also use normalization methods or domain shift approaches to deal with such differences.

5 Conclusion and future work

Early brain tumor detection and identification can greatly lower the death rates linked with these diseases. Different approaches to brain tumor classification depend on magnetic resonance imaging (MRI), which is indispensable. This paper offers a framework for the multi-classification of brain tumors based on MRI scans from affected patients. Four main kinds of brain cancers were classified using the DenseNet121 model. Finding

and classifying tumors is quite difficult due to different brain tissue characteristics, tumor location, size, form, uneven or unclear borders, and changes in severity across patients. To handle this complexity, we needed an advanced methodology that combined machine learning and deep learning methods. The study made use of pre-trained models distinguished for their flexibility over many datasets. With 96.90%, DenseNet121 showed the best performance among all the traditional classifiers studied in this work. The outcomes clearly improved upon current classification techniques. The results greatly help radiologists and neurologists make accurate diagnoses and treatment decisions; hence, they support computer-aided diagnosis. Using the suggested approach, we noted encouraging results that emphasize its future relevance in MRI-based brain tumor segmentation. Even though the given model showed promising performance on the Kaggle dataset, in the future, the model will be validated on external data, including BRATS 2024 or clinical MRI data. This validation will assist in determining the strength of the model in diverse patient groups and imaging settings, which is paramount to clinical implementation. Although employing several designs might improve accuracy even more, this work is limited to using a single model.

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Author contributions

MR : Conceptualization, Methodology, Validation, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization. MAJ : Methodology, Investigation, Software, Writing – review & editing, Data curation, AA : Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – review & editing, Visualization. JR : Validation, Formal analysis, Writing – original draft, Writing – review & editing. T.A.N. A , Al and NS : Investigation, Resources, Writing – original draft, Project administration .

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Data availability

The authors confirm that the supporting data and findings of this study is available on request to the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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